

ActInf GuestStream #032.1 ~ Adam Pease "A Neuro-Symbolic Approach to Language Understanding"

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hello and welcome this is octave guest stream number 32.1 it's January 10 2023 we're here with Adam peace and we will be hearing a presentation followed by a discussion so Adam thank you very much for joining today looking forward to this and off to you okay thanks very much for having me uh looking forward to presenting a few bits of initial research as well as some background on long-term ongoing Research into understanding language and trying to mimic some just some elements of human thought everything I do is open source so I've got some links for you there I also have a pretty active YouTube channel although it's been a little quiet lately so there's a lot of material here that I won't be able to go into in depth and so I'm eager to answer questions when we get to the question period and also would encourage folks to take a look at that channel where there's longer video format presentations about a whole host of topics that we'll probably touch on today so I have a large ontology I'll explain in a minute what I mean by an ontology that's called Sumo uh and I'm also going to reference some work that Facebook has done called their baby Corpus so that's the background for the cartoons that I'm showing you the main topic that I want to cover today is a notion of Common Sense Auto formalization my objective is to be able to make statements in human natural language just in English at the moment uh ask questions also in natural language and get answers but get answers with explanation something that uh modern question answering systems that we hear about each day in the popular press are generally not capable of doing you might get an answer but you don't know why and knowing why and understanding and explanation is very important so we're not talking about just information retrieval we're not talking about Google where you ask a question and you get a list of documents in which there might be an answer or even Google's occasional capabilities for very specific sorts of questions where they give you an answer but they'll never tell you why it's more like you know you ask what's the weather today you get a nice concise link to the weather but you won't get an explanation for why the weather is what it is so what do I mean by ontology are

you meaning to share we see the first slide you're still only seeing the first slide okay I'm on slide three so let's go back to I'm not sure why that's no interest showing up but uh good it looks good now Common Sense we'll stick with this format then instead of the slide presentation format thank you it looks good okay so here's what I just talked about um so let me give you my definition for what is an ontology because this is a word that at a certain point in recent history maybe about a decade or two ago got overloaded so much now that just about anything is called an ontology and so for me to distinguish ontology from things we already had words for like semantic Network or schema I consider an ontology to be essentially a dictionary for computers to read and that also requires a bit of unpacking what is a dictionary for computers to read well for me it means minimally we've got to have terms a set of terms that label things in our world but most importantly we have to have a set of definitions that are stated in a computable language so just stating things in English or French that are definitions for words well that's a dictionary it's not an ontology and if we don't have definitions in a some sort of mathematical language or programming language of some sort then our machines are only going to have the barest abilities to manipulate or do things with those natural language definitions so I think they have to be in a formal language to nail down meanings that we have shared meaning so we have shared meaning between people that that agree to use a particular ontology and maybe more importantly we have shared meaning between people and computers so that the same constraints the same inferences that people would be able to make with these terms the computer is also able to make because of these mathematically expressed definitions so that's what I've spent the last 22 years trying to build up and this is work that draws on Linguistics philosophy and computer science as well as logic as a sub-discipline of of probably all of those three 3. and one reason I'm doing this nowadays artificial intelligence has become almost synonymous with machine learning so I need to just to explain a little bit well why am I doing this this is sort of old-fashioned AI of writing down lots of rules why do we still need to do this well I'm a big fan of Daniel Kahneman's work and this is just one justification of many but hopefully because it's very well articulated in his work I hopefully this can be at least the start of a justification for some of you who might be unfamiliar with the this symbolic approach that used to be the primary approach in artificial intelligence I think that he makes a good case that humans have two modes of thought we have an instinctive intuitive sensory pattern matching kind of process that's good for face or voice recognition or making a very snap judgment about what to do in a particular situation uh that is you know learned from the presentation of many many similar situations and that sort

of mode of thought fits very well with statistically based machine learning neural network steep neural networks and so forth that's an important part of intelligence and it's led to a lot of important uh commercial activities but there is another mode of intelligence that uh Professor Kahneman I think illustrates quite clearly that is a slower more deliberative deductive approach where people can explain what they do as opposed to just give a reaction and it seems like in large parts of the AI Community people are starting to recognize that yes there is still this this set of things that we require our computers to do that they're not doing very well with the current neuros neural approach um Merrick approach that we need another mode of tool set for and so we're starting to see this keyword neurosymbolic approach of trying to balance uh statistical machine learning approaches with some sort of explicit representation in a symbolic form and it's this latter bit that I've been doing a lot of work with and I think it's the combination of these two of main approaches that's going to lead us to some more exciting capability that's just my assessment of the state of the field but there is some emerging evidence that this in fact is the case that we have a potential with integrating these two approaches to handle a broader range of things that would look a lot more like human along with this Grand goal of doing AI I've also been applying uh this work in creating this large dictionary to some far more prosaic stuff in data integration the bulk of the commercial Consulting work that I've been doing really over the past 10 years is just looking at a software engineering methodology of trying to capture the formal meaning of things that we use in computer science databases spreadsheets and so forth you know anytime you have a spreadsheet you usually have a column a set of columns across the top of your spreadsheet that has some labels and it's very unusual for people to have any sort of explicit definitions of what those labels actually mean it's usually just assumed that people by looking at those labels you know if you're in the same Community or you can ask for an explanation that gives you a good sense of what are what's the intended semantics of that particular column based on that particular label and what I found is that you know it might work for a spreadsheet where you've got a few dozen columns and you're working closely with with some long time colleagues but as soon as you get to a larger organization with a larger set of symbols labels where people may be working at different locations maybe they've never even met each other where there's uh uh interactions occurring over the space of years in large multinational corporations that people's initial assumptions about how clear their labels are are usually completely wrong and so people wind up using data in dramatically different ways inconsistent with its original intent precisely because nobody's taken the time to actually

Define things and even in the rare cases of organizations that do have data dictionaries that Define their terms the intended meaning of their terms they tend to Define them in very informal ways using natural language definitions which means then that the machines can do very little to help ensure that people's intended specification of semantics is consistent across a large body of terms you might be able to have a human being keep a data dictionary of 100 terms consistent and well-organized uh over time over a long period of time but once you get two thousands or tens of thousands of terms it's well beyond human capacity to keep such a product consistent you need Automation and unless you have uh definitions that are explicit and in a computable language the computer can do very little to help keep those definitions consistent so that's one of the big byproducts of all of this work so far and I use techniques like automated they're improving but you know I don't have time to talk about now but would love to talk about it in in detail for anyone who's interested to keep these definitions consistent so I have a logical approach for the semantics of terms lexical semantics the the semantics of the words we use and this extends also to sort of computer science constructs like labels for tables or columns or database elements and so forth right and this is useful anywhere pretty much anyone has a spreadsheet anyone has a database this is a something that uh people don't realize they could make use of people seem to accept that uh they're gonna need to have a lot of human to human to communication to explain the meanings of terms and there's just another way right and that's what I've developed and that's what I'm basing this work on language understanding on there's a common belief that oh this is just too hard right um that we should be able to come up with these things easily there's too much of them meanings change and so forth once you take the perspective that there is a need for uh precise semantics for our labels and that this precise semantics uh is something like definition definition in a dictionary but in a formal language then it's not just a sort of a graph or a list of relationships or likelihoods or or probabilities um it's it's starts to be hard to imagine then how a machine would just come up with this sort of thing automatically and of course we have another another venue in which we take this sort of harder approach of programming and don't expect machines to just do everything automatically and that's programming right we have many many millions of programmers in the world now it's they're expensive it is a time-consuming process to program things but we invest in this because there simply isn't another way to do it and we we make this Enterprise of programming feasible through a couple of of points one is having very rich languages so we can say everything that we want to say so we don't want to be restricted in the sort of programs we can write similarly we need to make sure we're not

restricted in the sorts of definitions that we can write and the way that we make that practical is through reuse your modern programmer if you deliver some software you're you're writing maybe one percent of the program one percent of the code that's delivered most of what you deliver is going to be a set of large libraries and reusable components operating system database a web server and any number of other things the Java collection classes numpy libraries if you're doing python the way in which modern software development is practical is through reuse and so that's the other thing that I've been working to provide is this thing called Sumo which is a library of definitions that can be reused that take this problem that many people would otherwise consider to be completely intractable manual creation of knowledge and makes it eminently practical because you're actually reusing most of your semantics by virtue of using this Library right so this uh Sumo like sumo wrestler suggested upper merged ontology started in the year 2000 it started as just an upper ontology so the the acronym is a bit of a misnomer a bit of an anachronism if you will right now that it's really a comprehensive ontology not just an upper ontology because it's grown to be about 20 000 terms eighty thousand uh handwritten statements in an expressive higher order mathematical logic it includes links to vast fact bases like the Iago system lots of translations from its authored format into some of the standard representations used in the automated theorem to proven community and as I mentioned it's all open source and online and every version has been open source there's nothing that's hailed back for commercial reasons uh let me mention wordnet so another common misconception in ontology is that we have to use words and follow the definitions of words and as as we should all know words are ambiguous um and words are polysemous the same word can have multiple meanings so we've created a very uh clear division but also a relationship between the senses the concepts that we have in the world that have formal definitions and the labels that we as humans put on those Concepts to communicate them to each other and this also makes it easy to ensure that ontology is not tied to any particular one language because as we any one of us who's who's multilingual knows an English word and a Tagalog word can have the same underlying semantics the same underlying real world referent but we use different labels and so we need to make sure that we can have vocabularies that are mapped to our common single formal ontology that tries to explain and Define Concepts in our world and not let sort of language and labels and intuitions about those labels creep into our formal definition so we reuse Princeton's wordnet which is a project going on for about 30 years now I think and it has also projections into multiple languages polish word net Tagalog wordnet for example uh here is uh there's continuing work

on that from uh which I'm very excited because I started a Tagalog word net back uh in the 2000s at De La Salle University and so it's great to see that that assertive effort continuing there's word nets for Arabic and Chinese and you know many other languages and so keep those labels in English those dictionaries human dictionaries distinct but related to the formal inventory of mathematically expressed Concepts and so back in the early 2000s we took an effort to go through all 117 000 sin sets in wordnet all 117 000 linguistically labeled Concepts and map them all to Sumo one at a time by hand it's just an enormous effort where we couldn't have really done it automatically it required human inspection we invested that effort and it since paid off for the linguistic work that we do with this ontology why do we need both well the formal definitions are things that just aren't in wordnet or any other dictionary you might have a word for earlier and a natural language definition for what it means for some events to be earlier than another but we need a precise mathematical logical definition of what earlier means and so Sumo has this and many many other axioms that individually are trivial but together add up to creating a certain understanding about the world that without it machines just don't have we need a machine to know that if one event is earlier than another that the end point of this first event is before the beginning point of the second event so we've got a formal mathematical definition of earlier which you just don't have in a dictionary this allows machine to do logic inference automated inference and and give you conclusions and explain those conclusions not a criticism of wordnet just they're different products with different needs and we need both also I get a lot of uh of questions about what kind of logic do we need do we really need a very complicated expressive logic it's very popular nowadays for people to use taxonomies or knowledge graphs well knowledge graphs are essentially semantic networks it's a knowledge representation that was created back really in the 1960s and so it does surprise me that it's kind of you know what's old is New Again in artificial intelligence somehow and that that knowledge graphs are a big thing now but it's a knowledge representation technology that leaves out a lot of stuff that we would need to say about the real world in order to define the terms that we use so natural language definitions aren't enough and we believe that defining our our terms or our Concepts is good which particular language do we choose and I've had people often say you know with usually with really without any backup oh those are the rare cases well I got frustrated with that so I did a paper recently right well now almost two years old about trying to quantify how often do we use constructs in language that require more than just a graph of relationships and so I came up with some statistics I don't I'm not going to go into this paper in great detail but if you're interested I'd love to follow up with you

and I came up with that in two uh looking at two large corpora Brown Corpus is kind of old and kind of small by modern standards but I also use the Corpus of Contemporary American English took a random sample of sentences uh created an automated system for looking at some various features that require in language a translation into using an expressive logic such as modal Expressions can May should might or expressions of authorship you know said wrote epistemic knows believes these are words in language that trigger the need for a more expressive representation we can also I could give you a lot of detail on why that's directly entailed this is not just a you know this is not just my opinion trying to to Advocate a more expressive of logic these are very sort of strict things that are well known in linguistics but at least now it's well Quantified that about 50 percent of statements taken from a random sample of a large Corpus do actually require these more expressive semantics so I think I've made a strong case that we can't just get away with graphs unless we want to leave out about half of the things that people say to each other and that tells me that that we really need this as more expressive logic okay so so that's all the Preamble what is that was the the library of semantics of meaning that I already have and how can I use that now to try to create a system and this is research right and it's uh relatively experimental even at the moment that the attempt is to try to create a system where the machine can translate automatically from English sentences into a formal precise semantics using these terms out of the ontology that themselves are precisely defined so I have the one at the starting point uh English statements or questions I translate these things to logic I want to be able to send them to a powerful theorem prover that's at least first order logic plus equality but preferably higher order logic so I can handle some of these modal Expressions there are a few such provers and I want it to engage in its process of doing theorem proving not only with the statements and the question that a particular user has provided at one time but also this Rich background that describes the world so that the machine is not just doing you know a sort of simple matching even with these powerful tools but it's really taking into account background knowledge World Knowledge in the same way that a that a person would and then when it's done when it's got an answer I want it to be able to explain what it does and Theorem provers have this or capability already it's built in they were designed to provide proofs because a lot of the work in fear improving is for fear improving in advanced mathematics so there's a quiet but significant community of mathematicians and computer scientists that are using automated they're improving to prove novel theorems in mathematics and because they're in mathematics they have a very strict standard of what constitutes a real work you have to show a proof just like almost all of us

probably experienced in high school geometry of proving one triangle is the same as another through maybe the side angle side process machines can now do this and much more significant and complicated and sophisticated sorts of proofs and provide detailed uh breakdown of how it reached its conclusion and I take this as to being something that's required if you ask another person what's the answer to question X okay they give you y as an answer but then you want to know yeah what's the detail how did you come to that conclusion it's not just enough to say well I told you so so believe me sometimes you might get away with that but not all the time especially not for anything that's that's complicated and so our machine should be able to do that as well so I've done a number of forays into this Grand goal over the course of my career I started out with something called controlled English to logic translation where I had a hand-built grammar of a small subset of English that was designed to be unambiguous it was all written in prologue and I was able to create expressions using Sumo terms in its logic which was mostly first order logic back then in the early 2000s and it was you know somewhat somewhat successful but it was a fairly restricted subset of English and so the parser would break and say syntax error all too often and that was kind of frustrating um it was sort of like what one experiences today with something like Alexa or or Google where you ask a question says I'm sorry I don't understand right and so people wind up self self governing their uh their uh utterances to adapt to the machine and people were able to do that with my kelp system but it's still kind of frustrating I'd like to be able to handle text that was written not for that particular system and not dumbed down on the fly by a human but rather any arbitrary sort of text so then I started working with Stanford's core NLP system and using dependency parse uh Transformations and trying to turn dependency parses into logic I did have some success with that but I found that even as good as machine learning based statistical parsers are there's still a lot of cases where they bring breakdown in fact they they give a lot of metrics that look very encouraging usually on a per word or per symbol basis and they'll say or it were 97 percent correct on uh unlabeled attachment score is one that's used in the dependency parsing community and that sounds like a very high percentage but in a your average sentence in English is depending on you know who you talk to what uh how you how you collect the statistics about 24 words well if you look at uh 97 percent raised to the 24th power um then you get a problem and you get actually less than 50 of your sentences are going to be parsed correctly and again that's unlabeled attachment score for dependency parses that merely means that two words are found to have the same dependency relation without a commitment to what exactly that relation is well you need the label Detachment score which

of course is lower and you need it to be correct for all of these words to every really a coherent understanding of a sentence if you're going to do any sort of automatic translation so although this the systems are good and they've certainly gotten better from 2015 I mean Stanford stanza is is yet another you know a few percentage points better than core NLP was in terms of its dependency parsing but it's still not reliable enough not consistent not stable enough uh for me to use that as a basis for translation and then there was all the problem of well how do I actually take that that output and reliably create logic from it and that meant at that time I was doing a hand-built rule base and it just didn't work in the limit it was encouraging I think there's potential if you had enough people working on it a bit more research but it wasn't the way in which I thought I could reach success so I'm doing something different now so I'm I'm doing what I guess is more prevalent now in machine learning than instead of doing a pipeline of machine learning operations trying to do everything all at once with extraordinarily large corpora uh where I am trying to train a system uh to do auto formalization of language to go straight from language to logic without any intermediate approach like dependency parsing and this is based on some of the success uh with my colleague Joseph Urban who's been doing this for sort of work for mathematics so one of the challenges of using uh Fair improvers automated theorem provers for mathematics is that as formal as mathematics is it's not really quite formal enough in a mathematics textbook there's still a lot of text and there's still a lot of things that aren't themselves precisely defined and there's also a lot of stuff there in mathematics that's not in a common and computable format so for a long time there's been this problem of how do we take mathematical textbooks and turn them into extraordinarily detailed formalizations that an automated theorem prover can handle in first order logic plus equality or some other kind of mathematical logic so Joseph has had considerable success at this and we decided to team up on trying to do an even this even harder problem of just looking at unrestricted texts not just mathematics and they've been using uh Google's neural machine translation system which is built on top of tensorflow where at the moment using kind of a relatively old version of that we tried a newer version it turned out to actually be slower for our purposes but uh we need to do a lot more experimentation and make this work more popular so people tell us where we're going wrong and maybe there's some better machine learning libraries that we could use for this so this is the the architecture now I want to tell you a bit more about how we actually come up with this set of natural language logic pairs because you know like other work of this sort the libraries of pairs of your input output pairs that you're trying to train has to be really really big you know orders of magnitude bigger than any

sort of pipeline approach because you're trying to tackle the whole problem hand-hand and there's a lot more variability your statistical significance for the appearance of any given feature correlated with any other given feature is really low he has a very long tail in language of things that are not seen too often so your Corpus has to be very big how do we get that Corpus we certainly not feasible to translate a lot of English by hand into logic so how do we get this thing to begin with so we were aware of this project done at Facebook called the baby Corpus in part because uh tomas mikolov has returned to his home country of the Czech Republic and now works with Yosef Urban at CTU Prague so we had a benefit of getting some of his perspective on this work it was basically an effort to try to teach machines how to do simple inferences and we'll look at some of those simple inferences in a moment and in order to do that they tried to generate a very large synthetic data set of natural simple natural language sentences and simple natural language inferences that they wanted the machines to be able to train up to handle um this ultimately they used really only a few sort of rules or patterns I think tomasia said there's something like 20 different patterns I've looked at the code it's written in Lua it wouldn't be hard to rewrite it in a sort of more more common language I think it's not particularly complicated code not a huge code base a lot of simple patterns a great idea but I think we can do better at it given the resources that Sumo provides for Generation as well as understanding and anchoring meaning right so they had a lot of their knowledge representation such as it is was things like you know you have a bedroom a thing a location actors uh existing in this world it sort of harks back to uh the blocks world of uh Patrick Henry Winston uh I think it was Winston hope I've got that reference right you know back in the 70s uh there's this notion that we could do uh there's a system called uh sure glue I think it was that did operations in a world of blocks so a constrained World there were actors there were a few things that could be manipulated and the system was able to understand very simple utterances well they've kind of turned this on its head created a knowledge representation for a certain world and then tried to generate a lot of sentences from it to create this synthetic model of things that a machine should be able to do inferences about and so here's some of the content of their questions so Fred is either in the school or the park uh Mary went back to the office is Mary in the office so the first sentence is just a confusion uh uh element of of the the Corpus it's really if Mary went back to the office yes she's in the office uh bill is either in the kitchen or the park another confusion sense Fred moved to the cinema it's Fred in the park well no he's not in the park but park has been seen as a token earlier and so they wanted to make sure that the automated learning system isn't just con uh trying to find

proximity is a good thing to include in their data set so lots of stuff like this and more complicated things also just presented really some of the most simple examples but all generated from a very simple knowledge model Concepts that weren't really defined and a massive presentation of tiny very mind-numbingly tiny variations in a wide variety of sentences so we started out thinking well let's just cover this the stuff that's in the baby Corpus so we understand what they're doing and the big advantage that we thought we could bring to this is we don't have to learn logic you know in a way they were they had a simple Corpus that was trying to accomplish something that's very hard which was to teach machines just through massive examples how to do logical inferences or a certain subset of logical inferences we have an easier problem if we use a fear improver we don't have to teach of the system to do logical inference we already have that we just need to get the knowledge representation from English into logic and then let the theorem provers do their thing so again very simple World they had objects locations actors States actions a few other decorations and so we tried to figure out really what they meant in a way that was hard because they didn't have a distinction between instances and classes so uh you know we can talk about in Sumo relationship that in an action one thing is an object transferred from one location to another they have this notion of being gettable which is really a modal and so we had to figure out what what did they mean they didn't really mean a modal uh they meant yeah some something that can be transferred and then there has to be some kind of transferred example we wound up interpreting a lot of their symbols not as classes so the notion of an apple is a class but the Apple it gets transferred in a particular sentence is an instance so this difference between existential and Universal quantifications so we'd create a term like apple one is a kind of apple is an instance of an apple right and that's a gettable thing and it's located in a particular area Etc so I don't have a lot of time to go into our knowledge representation but uh if you want a full course on Sumo I can do that at some point and there's a lot of videos online also so maybe I'll skip through a little bit of this uh we wanted to make sure we covered you know all of their stuff uh you know we did they they referred to things like chocolate and milk and garden and container and box and we actually had all this in Sumo so getting back to this notion of well if you had to Define all this stuff from scratch you'd be tempted just like the Facebook guys did to just use the labels and assume everything everybody means the same thing but we didn't have to do that we already had these terms precisely defined in Sumo and we're able just to reuse them so we did a big table this is just a portion of that table of all the different things that they referred to that we already had or in some a few cases didn't have and we needed to add them but even you

know adding the new Concepts and their definitions easy because the new Concepts had definitions that were based on in turn other terms we already had so we just had to combine them in the right ways you know we we didn't take for granted uh the that we were just using the first meaning of things for example they had the word pajamas in some of their database well pajamas have a couple of meanings in the dictionary yeah I wasn't all that familiar with the second meaning but it does exist and so we want to make sure we're using the right meaning so we also did mappings to wordnet of the terms we defined so that we had disambiguation potential for our natural language uh there are a number of things because of this uh small and artificial World they they had uh inferences like if you're hungry uh then essentially you go to the kitchen and eat something that's in the kitchen that doesn't really hold in the real world obviously we're don't we're not in this fantasy constrained world that uh hunger immediately entails you go to the kitchen and eat an apple um but so we had to create some kind of some bogus rules in Sumo that were very specific just to this Corpus to try to replicate the inferences that they were doing so here's a here's the rule that gives the right answer for this Corpus uh it gets a little complicated to State it but we can state it in Sumo that if you have a cognitive agent that's hungry and that's at a particular time then there's a translocation event uh or that is desired by that agent and that translocation has a destination of the kitchen where there's a getting a of an apple okay a rule that that doesn't belong to be in Sumo uh on any permanent basis but is what's used to cover the semantics of this particular Corpus and we looked at things like being bored and tired which they had they all say if you're bored and tired you go to sleep okay that's I wish that that were true for all of us hard-working Engineers that we had that option we don't at all times but that was what covered the semantics of the Corpus so we had to do similar things for bored and tired um they don't really have weights and measures and units in the baby ontology Sumo of course has detailed ontology of weights and measures of units all the system international measures and we had to figure out how we were going to translate these hacks that were in this Corpus of you know things have a numerical size one two or three something like that how are we going to represent that so we figured out a similar hack for covering that all right so you know we we did this exploration to make sure that uh we could cover one of these sample worlds but what we really want to do is generate a much larger corpus with a more interesting more varied set of things that are referred to in the real world so we wanted to do uh a synthetic text generation as our first step so that we kind of seed this system long term we know that we need to go beyond that synthetic test we're not there yet we're just still in this in the process of

creating this very large set of synthetic sentences because we can start with an idea or a knowledge representation about what we want to express in the world and generate logic syntactically and semantically valid logic expressions from that while at the same time because we've created the the state of the world artificially in a program we can then also create a sentence that models that state of the world so we go kind of both both directions both to logic and language from a very ad hoc sort of expression in some Java code about what's true in the real world connect allows us to do a lot of variation but it's never going to cover the full expressivity of natural language but we think it will cover enough that it gives us a head start where we can then start seeing well where does our automatic language to logic system break down what are the sentences where it fails where it generates something that's not even syntactically correct maybe or Circa or violates constraints that we know from sumo about how the world works and then try to add those as new patterns to our generation system so we can find one sentence that causes a problem we look at the area the the core issue of what causes the problem and look at you know maybe the things that are variable this sentence so we might have a sentence that's problematic that has some sub subject and an object where the subjects and the objects have a wide varieties that could things that could be substituted in that we can turn into sort of a template to try to cover another your pattern if you will in natural language and one of our possibilities I think will be to look to add framenet that is exactly such a large inventory of of these things you know I've I've wanted to do a projection to frame that into Sumo for for several decades now uh did some early work with uh the team there and we we published uh some some work about that in the mid 2000s but we didn't do a comprehensive inventory of all of their different frames and so I think there's a good project there uh to to do a more detailed exploration of all the frames and frame that and translating them to Sumo because the frames and frame it don't have a precise logical semantics which is one of the challenges I think that people have found with using that resource it's not anchored in the same way that it needs to be I think to be used more productively so so there's I think an interesting project there that would have a very practical outcome but for now I'm just going to talk about the synthetic text is because we're not even into that phase yet the synthetic text is the tough stuff we just started this work uh sort of mid-fall uh so that kind of gives you your perspective on on where we are this is going to be a long-term thing so I tried to collect all of the things that I thought I could detail both linguistically and in Sumo so things like uh propositional attitudes modals tenses names and rules and basic forms of transitive intransitive ditransitive verbs in this subject verb object basic a grammar of English oh try to say what

some of these things are so I'm going to give you an example it's it's a real example of stuff we can generate and it looks kind of funny because and I'll tell you why it looks funny and what we what we can do to fix it and uh it's other examples like it um but I'm you know giving you words and all of this extremely early uh research even though I think it's very promising I'm going to tell you what's wrong so far in the examples show stuff that that look kind of funny so you know an internet user traps a crustacean that's a pretty funny sentence it's syntactically valid and I think you can imagine an interpretation of that sentence that's legitimate in the real world okay I'm an internet user and I go down to the beach and I catch a crab you can imagine that actually happening but I'm not sure you'd ever find that sentence uttered by a person we think we can do better constraining ultimately our language generation through using large language models uh it should be in principle fairly easy to do so that we we generate stuff that's just a little bit more reasonable a little more common a little more expected um but this is what we've got right now because we're just creating these templates and we substitute in all of the the legitimate possibilities were legitimate it's just you know does it meet syntactic and type constraints specified by sumo and so it's possible for an interview user to trap a Crustaceans so we generate this sort of thing amongst many other possibilities and at the same time we generate the logic expression that you see on the left I'm sorry I'm throwing code at you and I don't have uh the the time here to go into the full syntax of this expression with the moment this is just first order logic so it's an existential quantifier saying there exists an internet user a trapping or an ambush is the term that's used in Sumo again remember labels and definitions are different so we used a particular software identifier of Ambush but its definition fits with what you would expect of trapping okay and just the word that we happen to be using in English is trapping the software programming language identifier we're using in the Sumo library is Ambush okay so you shouldn't expect those to be identical because we have different labels for things that we may have use consistently in terms of their formal semantics but their labels may vary so we have an agent that does something that agent is the internet user the object of the action is the crustacean um and the yeah the object of the action is something and that something is a crustacean all right so very simple sentence very simple logic formalization and so we generate tons of sentences like this for different objects and different agents and different actions this is your most most simple subject verb object sentence and we have lots of different objects that can play roles different agents that can play roles and different processes that are already defined in Sumo so we just go through the entire inventory and we also probabilistically generate that we look at uh each of these symbols in

Sumo has a connection to wordnet and wordnet has word frequencies for each of those uh those Concepts so by looking into wordnet we can see which Concepts formally Define Concepts in Sumo are more likely to occur in free text individually and so we do prioritize Our Generation on the basis of trying to make it more likely using a randomizer in Java and then a frequency table we try to make it more likely that we generate sentences with things that are more commonly seen But as we can see the the even though internet user may be fairly common um in in natural language text you wouldn't likely see it paired with trapping or a crustacean and so it's these pairwise selectional restrictions that we don't have right now which is why you get these funny combinations uh we can generate times uh so we can have the internet user trap station or trap decoration or will trap or is trapping and its formal temporal logic expression one thing you often find in simpler knowledge Expressions is verbs used as predicates without tense so you might have in a Knowledge Graph internet user a relation trapping or traps of some sort and then crustacean right this would be a more typical expression in a graph and that doesn't really capture what the entire sentence is saying right it says that there is a temporal qualification to what this what is happening in the sentence is happening now right and so uh there's a begin function of this activity and an end function of this activity and now is between those two things you have a dike dick as we say in linguistics right something that's being referred to that gives semantics to the sentence that's only meaningful in the context of the speaker and The Listener here they're now Etc there's a lot of these things and we've got to be able to handle that in order to capture the semantics in a sentence so it's not enough to say there's a trapping relation between the internet user and the crustacean we really have to go to a temporal modal logic Beyond first order logic with equality to capture this information and capture it in a way that's going to work not just now for this one isolated case but for all possible cases of these sort of temporal Expressions it's also common for people that don't have a background in modal logic to sort of assume that there's some kind of a fudge for this in a graph representation there isn't right and this is why logicians work on this stuff it's hard getting it right has taken many many decades of effort and so you can fudge it but essentially somewhere down the line you're going to get completely wrong conclusions nonsensical conclusions because you don't have this same strength of semantics in your representation that's why we need this difficult and expressive logic we have negation negation is another big problem for graphs right you've got to be able to negate an arbitrary conjunction or disjunction of other statements to do that you've got to have a construct that is beyond a particular relation that is is a sort of a contextualized relation it can

handle a whole expression logic already gives you this and getting negation right is really a necessity in logic or in any other representation you're trying to create so like you sometimes people try to say well I can do that in my graph or I can do that in my description logic and well maybe you can for this one point case but essentially if you're doing this in your ad hoc representation that's not a formal expressive logic you're going to wind up syntactically recreating what logicians have already done and done many years prior and you're going to sort of discover these things on a point by point basis instead of just getting them right from the start so encourage people to to you use a representation that can handle the full complexity of the semantics of English and only a higher order logic will give you that so here's case an internet user doesn't trap a crustacean on negating this entire complex existentially Quantified expression that includes a temporal modal logic expression within it what if I have you know an agent Patrick desires that the internet user trap a crustacean pretty common to see in news texts you know uh Kevin McCarthy said he believes that the next vote for speaker will be successful um you know bolsonaro desires that he uh be the leader of the country again these are very common uh expressions in natural language they are not infrequent and they're difficult and they require this sort of expressive logic and that's the sort of thing that we're handling we're able to generate this stuff and have a representation that handles it and then maybe if we're lucky have a theorem prover that can handle it I mean higher order logic theorem proving is still difficult and somewhat experimental and so that's a parallel line of research um but you know we're not sort of hand waving away this problem we're not imagining it doesn't exist or imagining that we can really handle all this complexity and language without uh this this additional set of computational support that we believe is you know in fact really required uh so here just making the logic and the sentence more and more difficult the plumber knows that Patrick desires that the internet user traps a crustacean you know pretty artificial sounding sentence admittedly but not all such constructs uh for sentences are uh that that rare right you do get in a news report um The New York Times reported that uh the speaker desires that this bill pass without further amendment that would not be a particularly unusual sentence and yet it follows exactly this pattern so one of the things that we need to do and if anyone is at all intrigued by this work and thinking well how can I uh contribute which would be wonderful uh here's a great way to do that one thing that is that we need that needs to be added to sumo and yet is also fairly well constrained in terms of the the complexity of the underlying axioms that need to be expressed so it's possible to make these fairly formulaic and in fact the use that we make of them inside my Java code that

generates logic language pairs is very formulaic and will look for only uh axioms that that are of a very particular restricted form in order to do its restriction of which sentences get generated we have this notion of capability a capability relation at Sumo so it says that uh some kind of thing is capable of Performing some kind of role in some kind of action so ears are capable of hearing guns are capable of shooting and we have to talk about things that might happen in order to restrict the sentences that we generate we'd like to be able to say that John you know John it can't be eating a boat dock we might generate something like that now but we want to be able to restrict eating to have an object transferred or a resource that is only food ideally food for that particular animal so we want termites to be able to eat Boat Docks but we don't want Bob the plumber to be able to eat a boat tuck uh and we have a lot of processes and we have a lot of objects in Sumo and so creating an inventory of these different capability Expressions would eliminate I think a lot of the kind of funny sounding sentences we generate now that said I think it's not catastrophic that we have some of these funny sentences because remember that we're ultimately trying to go from uh organically occurring natural language text in the wild in order into arbitrary logic Expressions so the fact that we have other sentences that kind of have a similar form but don't really make a lot of common sense may not be a big issue because as long as we have enough examples of things that are reasonable that do cover the vocabulary we care about that do have reasonable selectional restrictions that that'll still us enable us to translate the good stuff that occurs the sensible sentences into sensible logic expressions and the fact that we have a corporate purpose of translations where the translation is reasonable but the core sentence is kind of focus may just be background noise we may be able to get away with that but even if that proves not to be a problem it still would be better for us to have in Sumo these capability Expressions just to add to our knowledge of how the world works so if anybody is intrigued by this I'd love to help you get started because I think it would be not too hard to do and would be of great value to the project so we have a lot of as I've said I've said a couple times now that we have a lot of processes we have a lot of objects well what what do I mean by a lot well here's uh just a segment of the natural language comments on a portion of the hierarchy starting from the top here are all the process types in Sumo we have 1300 of them at last count and here are a number of uh so in a lot of sentences we don't restrict what processes can hold for a given sentence and we'd like to have again these selectional restrictions based on capability expressions for the type of objects that can participate in the types of processes so we don't get silly stuff for each of these processes we have mappings to verbs and wordnet so that's what

allows us to cycle to iterate through all the process types and still generate a sentence not using the labels in Sumo but rather using the labels the legitimate words in the word net dictionary so for example for Ambush it's mapped to a particular numbered synset a collection of synonyms collection of synonymous words and wordnet that linguistically identifies a sense in English and so we have Ambush that's mapped to since set blah blah blah 926. uh it has a certain definition has a certain set of synonyms and buscade Ambush lying in weight and a trap and that's why we saw a trapping earlier the internet user traps a crustacean that was exactly this equivalent symbolt uh since that for ambush um we have lots of case rules so the way these are the ways in which entities can participate in events uh there are some simple systems including some simple systems in linguistics if you look at uh I think verbnets has some of this set of restrictions so agent patient instrument resource these are very common um and so you'll also see Agent Zero agent one sometimes as labels and semi-formal Linguistics talking about the roles that entities play in activities uh and so occasionally I've gotten this question which I think is funny is like well why isn't it enough to have just agent patient and instrument well yes that does cover at a high level maybe exhaustibly uh the roles that things can play I mean James pistiovsky also has this set of qualier relations this is a common uh I think approach in linguistics to try to explain language through a small number of covering predicates or abstractions and while that might be fine for abstract Linguistics in practice there's a lot more that we can say about the world about how things interact in events that happen and so we've come up with 67 of them and I think this is you know many orders of magnitude too small but it's a good start that's the stuff that we've found that's that's been useful over the years so invading virus Cadillac Catalyst broker enjoys reagent all of these relate Nations and many of them are sub relations of things like agent and patient and instrument and resource but they're more specific they they entail different additional knowledge when they're used and it's knowledge which fits with our knowledge of the we have lots of objects so here's just the corpuscular objects things that are not substances that have Parts uh and we have 930 of them a lot of them or content bearing objects or things that may have writing or symbols on them that are are common in our world as well as organic objects like organisms um there's a lot of these and they all play different roles in different uh here are some of the roles that uh human beings or agents can play you know you can have a certain profession you can be a plumber or a programmer and you can switch freely from one to the other this you know past weekend and maybe my toilet breaks and I have to be a plumber today I'm a programmer uh maybe tonight I will be teaching a course and so I'll be a

lecturer um these different roles are free and uh flexible and large and they also uh are temporarily qualified right you can play different roles at different times so it's not enough to say as wordnet unfortunately does as a linguistic product that a sub type of human is plumber well you can be a plumber one time and not be a plumber another time and that doesn't mean you cease to be a human it's a bad relation to have formally although it's a reasonable relation to have linguistically because you can do the substitution tests that linguists do of substituting uh say a super class for a subclass and it works in a lot of sentences but it doesn't work formally doesn't work in logic so we need to separate out these roles and say that they're transient temporally qualified rules that people can have so we can generate against bogus stuff like Robert kills a boat dock doesn't really make any sense it makes sense syntactically but we need these capability restrictions to handle them and a lot of times we do have restrictions like the rule below here um but that it's just I I can't afford to do theorem proving while I'm doing generation if I need to generate a one terabyte Corpus of language logic pairs I can't prove the truth of every sentence this would take you know many years if just to do one run of the generation system I've got to find a way to optimize that and the way to optimize it is just to look at capability Expressions even though the the following rule that I'm showing up here says that the killing has to have an object that's an organism and a boat dock is not an organism so each this would be trapped if I could do fear improving for everything I generate but I just don't have time so I need to use this more restrictive form that I can turn into a sort of a lookup table and and make run much faster okay so I need this following expression that uh if there's a capability of killing the uh killing event uh only the patient has to be an organism if it's the patient or the object of the killing event it must be an organism I also had the the question of how much is too much you know right now I am generally able to generate say a terabyte uh Corpus in the span of a couple of hours and then train on it uh training takes about 28 hours on a multi-gpu fairly beefy server um how much is too much I don't have a sense of that how much is it too little you know we're getting a good Fidelity you know we break things up in the standard uh uh crane test uh split um but and we get good metrics you know we get a perplexity score that's down around 1.01 1.02 uh it's one reasonable measure for the Fidelity of your deep neural network training approach um but is that good enough I I don't know yet we just need to do a lot more experiments so we have about a hundred names that we use so we have a corpus of of common human names we have about these 300 rules we have lots of different propositional ad attitudes Each of which could be negated we have modals like handmade should that are also formally expressed in our modal logic those in turn

could be negated we have 400 different sorts of roles like plumber and teacher 300 processes those in fast could be negated and we have past present future and then the progressive tense for each of those all those possibilities we've got 930 direct corpuscular objects and then we've got substances which I haven't listed on here we've got lots of substances like water or gasoline 67 case rules and again 930 of those same corpuscular objects that can be indirect objects and then I've also added a bunch of politeness expressions like can I may I should uh do the following so a lot of these modal Expressions can can be become politeness so please may I Etc so we can handle stuff in uh chatbot or conversational kind of stuff that's that's very commonly seen so I did an inventory of stuff that a lot of the chat Bots use for their sentence patterns to make sure we cover that but a lot of these combinations are nonsense so I've got to somehow pair it down if not if it's not a problem in terms of creating a lot of junk that can otherwise get ignored when I'm translating a sentence at the very least it slows down training and so as I make the Corpus larger I I have to Pare down the nonsense just so that my training phase is possible because I don't have the resources of open AI to spend three million dollars a month on running their you know chat GPT for the world right I've got to run on one server that my friend Prague has access to um so there's lots more that I can still do uh units and measures I've only begun starting to do that I've got a little bit of that so I can say you know uh you know Bob puts uh three three birds in his wagon uh uh or three uh pounds of steak in his jar something of that sort uh because we've got a whole set of I think they're about 60 different system international uh units and Sumo has them all and then of course your values can be within all sorts of different ranges we've got logical combinations of sentences that I could generate so Bob goes to the store hand Bob buys cookies um we can temporarily qualify all of these things I've got a little bit of that already so you know Bob goes to the store on Tuesday Bob goes to the store on January 27th of 2003. uh we have a bit of an issue that a lot of these case rules require different prepositions so I don't have those working correctly I need to actually do a whole set of patterns that are a little fragment of frame net if you will for uh different action types and which particular prepositions are historically used so you get in the car you get on the boat you go to the station I don't handle all of those correctly things we can do like with more than one agent right now agents are always singular in my examples we can certainly have Jack and Jill go up the hill and not just Jack I do have now just implemented quotes for said so we can handle the common this common form for news reports or novels or what have you lots more things that can be done to make this a more comprehensive Corpus long before I you know run out of stuff to do I I can cover a lot larger portion or subset of of

English um so here's some just some detail on how we run the model um I just got a script it's not too hard uh just takes a long time so future work uh once I get done exhausting the set of things I can think of that I can do then we need to start seeing where it fails on natural texts a big thing to do is to use either word to VEC or some large language model maybe as a way to Pare down the number of nonsensical combinations along with my preferred approach of having a lot more capability senses which will then add to the understanding the the competence of this core model regardless of whether we're trying to do language understanding or some other kind of project and I really want to do a comprehensive test on the the baby Corpus to just see if I can do better if I'm not learning and teaching the machine Learning System how to do inference but using a system that already knows how to do inference I think I should be able to get better scores but that remains a empirical question if I can do that I think maybe then people pay more attention to this work so it would be a nice demo okay that's that's it so thanks for learning about this early work as well as the more mature model that it's based on that's already being applied commercially and I would love it if I got somebody on this call so excited that you wanted to do some work together excellent thank you for the presentation Adam you can unshare or leave it up and we will begin with Dave providing some initial comments and questions yes one uh idea can you hear me now I can great one idea that occurred to me several points during your presentation was if you put part of the training a part of the pruning inside the modern by creating persistent agents with goals and beliefs Bob isn't just a transient pattern of activations but Bob is something that stays around for a while and perhaps Bob's desires evolve more slowly than Bob's beliefs including Bob he believes about Carol and Carol's relevance and whether Carol believes correct things about what he's interested in right now have you gotten her out of something as as Charming as that well that's what the baby folks did right they they've got this model World model and they generate sentences and variations of those sentences based on that world model and we could certainly re-implement what they've the simple system that they've done in Lua um in a more complicated blocks world if you will that uses all the entities and objects that are in Sumo that's something I'd like to do uh I can only okay reconnecting here foreign thank you we've got Daniel back I had to move to a backup um source for a strange reason but let us continue the conversation with uh no video and uh yes the last thing you added was the capacities and the opportunity to contribute there please feel free to pick up there or I can pick up there and take it in a different direction yeah David I was just uh talking a little bit further about uh maybe some other work that's going on in uh psychology and uh ontologies emotions

and I was just commenting how um Sumo already has an extensive ontology of emotions but of course it has ontologies of many other things so anybody that wanted to study a representation of emotion for use in Psychotherapy or any other reason we'd kind of get for free all of this other stuff defining a model of the rest which would I think make that work have a greater impact over time it'd be more readily adoptable in different contexts because it has this link to a whole host of other semantic defined entities yeah one of the things that would make it a lot easier too to to participate across groups is if it were easier to actually run Sigma several of us have tried to install and run it I don't know that anybody in the active inference uh group has ever succeeded uh you do put for the for the the appropriate environment you have some really nice videos and I've tried to follow those in every case it broke down with some warning like uh if you keep going your Windows system May self-destruct why don't you stop now uh so it's pretty scary we even tried to go to a um a vendor a provider of cloud services items like Sumo at all that provide rigor in how those different state spaces get translated and mapped to each other like short text to long text long text to short text short text to Mouse movement although those different mappings are implausible in multiple ways without a formal semantics the kind that you're describing yeah yeah I agree maybe I could say a word about about uncertain reasoning um because this is an area that I think a lot of people that are used to being in a probabilistic representation often have some misconceptions about what's possible in logic usually based on the fact that the the logic that most people are exposed to academically is going to be first order logic where you do have just truth values binary truth values right it's either true or false and people want to be able to represent things that are uncertain or likely or unlikely and therefore assume that uh that logic can't handle that and well it can it's just you have to go beyond first order logic and so that's exactly why we're in the realm of higher order logic and Sumo and have modal expressions we want to be able to say and can say in the logic that something is likely or that one thing is more likely than another or that because of such and such event something has become more likely so we have to make a projection that includes some temporal reasoning as well so logic is uh you know you could call it a non-classical logic right I find that a little uncomfortable because most non-classical Logics are things like you know four valued Logics or uh things that don't just have true and faults well motologies do still just have true and false but they are able to describe an entire situation sort of with all its possibilities and conditionals possible things like possible worlds which allow us to capture some of these same intuitions that a probabilistic representation like Bayesian reasoning can't is I think there's a to my mind a clear

advantage of the symbolic representation it it is possible of course uh to have large libraries of empirical data to have uh realistic probability assessments and say that you know such and such an event is 73 likely in this context but a lot of times when I see people using probabilistic or basian representations they're sort of using them mathematics is precise thus the source data is often not precise people are making a guess you know I guess that it's probably 70 percent likely that the following will happen then they encode that in their system and I think those things are better represented as symbolic preferences so this is more likely than that but we don't know that it's 73 likely versus 56 percent likely an additional problem I think often not maybe not in your case but in certainly in many cases with people they're doing numeric probabilities is that the numeric probabilities force a total order essentially um and you've got these appearance of of precision where 73 is more likely than 56 but in reality if that comes from anything less than statistically rigorous empirical data then it's just sort of a confidence Factor like was used in a lot of old style expert systems that that doesn't give you reliable results so none of those criticisms may actually apply to your situation I don't have enough detail about what you're doing but I would ask you to consider those points awesome I'll give a thought on that and then Dave happy day hear from it so for any given model architecture symbolically represented like a symbolic regression or a Bayesian graph constraining on that model structure which in the variational inference world we just call Q it's just the variational distribution the one that we are getting to choose the family that we're parameterizing it from so it's like our designer symbolic model and why the data come in so then free energy which is the imperative for minimization whether it's the real-time variational free energy happening right now or it's the planned or anticipated expected free energy in either of those cases for a given generative model given Q distribution and given set of data you might have a very high confidence in something that was very inadequate and so in that way it's like there are distributions of Scatter Plots that can confuse a least squares regression or you get some sort of misleading model outcome and it may even be invisible you may be able to look at Diagnostics of the linear regression and tell that some sort of error or aberration is happening but you also may not like a kind of unknown unknowns question and in the case of cognitive modeling these State spaces can be very large and they're often trained with sparse data so this problem of overfitting and conditional and dynamic relationships so that something can be like true and trained upon for 100 video frames and then something changes and then there's a non-linear change in what can happen next or how things unfold through time so that's kind of like the complexity of action and perhaps could you speak to how action

is represented or treated or how the concept of action selection and planning and strategy and implementation impact how did these features develop in Sumo Through Time I think there was more to that question it sounds like Daniel's maybe have still having some connectivity that was it just how did the concept of action evolve Through Time how does the concept yeah um let's say uh there was a lot packed into your question I'm not sure I I grasped most of it but let me let me just maybe make a I'll give myself a related question which is uh how do we handle the fact that sometimes a machine Learning Systems the mathematics has such linearities that we get nonsensical conclusions right I think that's a fairly common problem um so I think that the interaction between these two models of the world the statistically machine learning derived model and the symbolic model is really important so that our machines don't go off the rails as it were I'm using a tournament called cognitive guard rails and I think this applies well to people you know people have instinctive reactions to many things and we have at the same time a deliberative cognitive mind that constrains those reactions we might react emotionally and react badly to something that sort of triggers Us in some way but then we have another process in our head that says well you know we're in a social situation and so no I don't want to punch this person that's offended me even though I might be tempted to do so um if we have a self-driving car that sees an object in front of it my initial reaction might be to plow through it but the cost to doing so might be very high so maybe I want to Swerve or break gradually and having this interaction between these two models I think can address some of this problem that we see uh with non-linearities of modern machine learning systems that do you know work really well in a lot of common cases and then sometimes just Veer off into Infinity as it were and do something completely ridiculous yes I I appreciate that response I think it's exactly like you described with the overfitting to the meaning and going off the rails from a cognitive perspective is there's like the known unknowns as an example especially there's a University of Wyoming I think did a nice paper maybe quite a while back now probably at least five years back that you would say yes it's a cat or a dog because it was just training on sort of a coincidental uh feature of pixels that maybe approximated the fur of the animal and so it couldn't recognize a uh you know recognized a test pattern as being a cat and it couldn't recognize a simple cartoon drawing as a cat because it'd never seen anything like that and wasn't looking at the essential features of the and so machines are really great at finding correlations but that doesn't mean they're the right correlations I think that's at the root of a lot of things that we call uh uh over training or overfitting um but we never know really whether it's it's fit correctly or overfit I mean there's sure there's some some numeric measures

that at times can tell you okay we're we're bouncing out of this local minimum and and our our uh our Fidelity has gone down numerically but we don't know what that really means in respect to the common sense larger world that is represented by something that's much larger than any given test set that we might be training on so I think that having a sort of diff completely different representation of the world can help guard against some of this and so I thought of using machine learning and symbolic systems together where the symbolic system uh addresses problems in the input data right by because in faulty input data erroneous input data is is always a big problem how do we know we're training on the right stuff how do we know that if we're training on a uh a system to diagnose uh check fraud for example or a credit card fraud that we don't have uh some bogus data that's crept in here there that might have a spurious correlation we've got say zeroed out fields or a field of 99 it's really as a don't care field but it's in the data and there's no easy way for us to discover it if we've got terabytes of data wouldn't it be nice to have a a common sense system that could act like an intern a human intern working at warp speed looking over our all of our input data to make sure we've uh not included any inputs that are at odds with what we know about the real world and then on the output side making a few inferences in our symbolic system could help the overall computational system not make really dumb conclusions like oh let we don't know what that things is so let's run over it might be the conclusion that that a statistical system has but we'd like to have our our combined agent uh be able to trap things like that in the same way that a person wouldn't presumably punch somebody else who's offended him because they know that that's both illegal and immoral thanks Dave do you want to add or ask something yeah a couple of the other approaches that sometimes simplify exactly these same kind of problems that that Daniel and Adam have both been coming through uh one is fuzzy logic uh sometimes a fuzzy solution and a probability probabilistic solution agree numerically but in other situations they don't and in many of those cases the fuzzy answer turns out to be way back when Lefty's out exifying and reacting to situations and it wasn't until he started calling the logic fuzzy that people were beating on him he was just coming up with a great results another approach that that's been impressing me uh is uh David Deutsch and Kiera marletto's work in Constructor Theory where they start by asking what can be done and what can't be done and what does it mean to do something what does it mean to be able to construct they generalize the notion of uh catalysts for instance as as a general Concept in systems theory and um they just identified I'll know a lot of things that people spend a lot of brain power on is just simply being nonsensical you can't construct that therefore you're wasting your time if you try to work

out how it could be done would be done with the consequences where if it could be done it just can't be so go on to something interesting well I'm not familiar with the latter work but I am a bit familiar with fuzzy logic and maybe I should just mention it you know this has been another thing where people that are familiar with fuzzy logic wonder how you could do this in a non-fuzzy logic the the challenge then comes down to again uh people that aren't familiar with Logics Beyond first order a lot of things we might say about like you know John John is tall okay um and maybe John how do we how do we compute whether John is tall and we have this problem of well if in a traditional logic system there who would have the problem that okay if he's above six feet he's 12 he's below six feet he's tall therefore if he's you know five feet 11 and 7 8 inches he's not tall and of course we all know from a common sense Point standpoint that's ridiculous and so we'd want to use fuzzy logic but in fact what we should be able to say is that he's tall with respect to a particular group right and if we can make these more complex statements generally involving sets of things and conditionals that are you know beyond first order logic we can express these things perfectly well in in this more expressive logic we don't always have to go to fuzzy logic because one of the challenges that I think if you have with fuzzy logic is again a lot of people that did work in this sort of world back you know like almost 30 years ago I think came up with things that were functioning well empirically as sort of hacks because they didn't have uh data about you know what is the actual statistical value curve of tallness it was just somebody's guess about what the curve of tallness membership should be for example so I'd like uh the the notion of fuzzy logic intuitively I'm not sure that I care for it so much on on a theoretical basis and on a practical basis there's simply another way to do it very interesting Dave want to add anything yeah now this is um if we have time for this this is we're a much deeper question about the way the Sumo rules are written is do we have time to go in in a lot of depth in this venue depth not necessarily breath not necessarily a big answer but okay I'll just ask this uh reading through the the rules a number of rules have conditions that very specifically asked in the first participant of a relation if the following holds then the following logic Applause if a different set of conditions on the paid recipient of action the indirect object or direct object of a process holds then you can apply the following logic and you know I look at that and I say wait a minute these are split these different rules are splitting off very different kind of real world situations within the same Sumo class well there's got to be two things have to be true in order to do that in order to have find these differentiating rules within a single class first you call those by the same name in whatever natural languages you were thinking in when you wrote

the rules and second the formal literally the abstract algebra of these various cases have to all be analogous they all have to be isomorphic and my name for that situation is you're covering analogical cases in the same Sumo class and in that sense the Sumo class itself is an analogy it's an analogy among these various cases which in different domains of apparition am I just not thinking discerningly enough about what a Sumo class represents or can assume across actually be a higher or analogy um there's it's hard for me to know because I mean there's no analogical reasoning happening so I'm not sure you're talking about analogies and there's no natural language happening once you're in the formal system so I'm not sure why you mentioned natural language I think we'd have to look at a very specific example and then you know we could we could run it on run run that example on a theorem pervert and you'd see exactly how things behave it's not something about you know classes per se um you know you just mentioned how how do rules work in a logic right you've got preconditions uh for the antecedent of a rule and you've got the consequent of a rule and yes indeed the the the symbols in that formal system have to match that's that's how logic works right I'll get some specific examples of an email empty okay sounds great in our last minutes here and also apologies for the internet disrupt um this is part of our long-term Journey learning about this and applying it to active inference so we're going to keep this thread alive and be able to bring some more informed uh areas of intersection and application but one that I wanted to mention was to return to your type 1 type 2 thinking and the way that there's a faster reflexive or habitual statistical type 1 inference and then a slower type 2 what you refer to as deliberate and deductive that involves symbolic logic and in chapter 5 of the active inference textbook there's a schematic of a dopaminergic neural circuit in the Central and peripheral nervous system where there's a dopamine related parameter computationally that's like a dial in the model that is controlling the balance between replicating actions according to the prior like just the Habit so it's some distribution of capacities affordances and then if there's three options it's 0.8 0.1 0.1 so then under the most habitual setting like what we might even call non-agentic or subconscious then that Vector of affordances is recapitulated in action selection and then in a different dopaminergic regime there is increasingly fine scale and potentially overfit application of the free energy and that can reflect like a symbolically informed deep generative model of the world which captures semantics of the world like nested time scales and temporal priority effects so it's interesting that the this area while deep hierarchical generative models often are described as if they have semantic value however that is usually just sort of specified or only explored in a few limited cases so integrating and working at that mapping between

the top down symbolic and the bottom-up data driven is going to be a really important interesting area so as a closing round or question what do you see in the ecosystem and or what directions if you didn't State them earlier you're working on this year well yeah the generative AI stuff certainly got a huge amount of press um it's it's fascinating as an intellectual exercise certainly and and it's fascinating to have a systems that can complete sentences and write stories and you know always have an answer to a question but as we're seeing with a lot of newer things that are that are being written as follow-ups to all the excitement that people are showing how well these these systems will give you an answer but it's often a complete hallucination right I saw one recently of saying well you know Hillary Clinton was the 43rd president well it came up with that somehow by merging together various texts that other people have written but it doesn't make sense together it's not true right um and so I I think these systems are interesting in the way that a a parrot can have a very limited conversation you know you can teach your parrot how to say Polly want a cracker and say thank you when you give it a cracker that doesn't mean it actually understands anything about what it's doing it's just a stimulus response um now first there is the philosophical question if the stimulus response uh pairing is complicated enough well isn't aren't human beings also just collections of stimulus and response so that that may be true but I don't think we're at that level of complexity yet for our machines and I think some of the things that we can get these generative models to do they're so obviously nonsensical human or good evidence that uh they're they're certainly not uh sentient and they're not even smart and they're certainly not very reliable if we have uh very challenging or higher high criticality tasks to worry about I think they'll be useful in the same way that Google search is useful it doesn't always get the right answer it will universally get many wrong answers but it's very powerful and useful tool when you have a human being available to filter out all the nonsense and just uh use that as a filter for what what they'd otherwise have to do you know it's not feasible for a human being to answer question by reading all the books in the library but if Google can give you 10 books and 10 pages in those 10 books where your answer may be found it's done you a great service even though it's given you you know Winnie the Pooh along with the handbook of analytical or organic chemistry or something where you're at your actual answer is found so in that same way chat gpt might be very useful in breaking your writer's block to help write the summary for a proposal or a student essay but you certainly don't want to have it write the whole essay for you and hand it in because you'll probably fail um so I'm not quite as worried about uh high school or college essays as some commentators have have been I think it'll be a

useful partner and that's about all and we've got to understand these limitations and ideally pair it with a symbolic system that really knows something about the real world so that they can work together awesome Dave first and then Adam what are your closing thoughts or reflections I've enjoyed this very much uh we've got lots of work to to plow forward and do I'm glad there's going to be common work that uh we can become affordances for one another on thank you gentlemen well thank you Dave for you know good questions good comments and and Daniel as well and and for setting this up um I guess I would encourage you know what I've maybe encouraged in the past is a lot of the things that that we've talked about in these several conversations we've had are General right uh you know we're saying what's an overarching direction for the field or something and I think that if we got down into some real specific examples and actually doing some representation together um then a lot of these uh more philosophical Notions would go away and we'd get down to doing something that has has practical utility that's why I also encourage uh actually having you know if people do say some of the Sumo exercises together because I think gave some of the questions that you were having over you know what what is the rule how is it working um I can answer all those questions and give you some very precise experience that would give you a a real actual understanding of how logic works and how Sumo Works uh if we took the time together to go through those exercises do some work with the theorem improver and it would clear up I think a lot of maybe misunderstandings or misconceptions and and give us a real path to do something concrete together because all the Talking is is great fun it's the the doing and writing of practical computational system that hopefully we'll all want to get to yes and say let us descend into the particular yes that's great I love it great ending the Stream